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ScienceDirect

Procedia Computer Science 270 (2025) 446–455

Procedia Computer Science

www.elsevier.com/locate/procedia

29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025)

Medformer: A Multitask Multimodal Foundational Model for Medical Imaging

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Abstract

Medical imaging datasets vary widely in modality, dimensionality, and clinical tasks. This diversity typically necessitates single-purpose deep learning models for each domain. We propose *Medformer*, a unified foundation model for multitask and multimodal medical imaging based on transformer architectures. Medformer uses two specialized modules called *Adaptformers*—one for input adaptation and one for output adaptation—along with learnable latent embeddings that encode dimensionality (2D vs. 3D), imaging modality (CT, X-ray, microscopy), anatomical region, and task requirements (classification, ordinal regression, etc.). These embeddings guide a shared transformer backbone, enabling broad parameter sharing across heterogeneous tasks. Experiments on the *MedMNIST* collection, comprising 18 diverse 2D/3D datasets, demonstrate that Medformer can match specialized baselines, particularly for data-scarce tasks, via both multi-task training and self-supervised pretraining. Results suggest that Medformer can serve as a flexible foundation model to unify disparate medical imaging domains within a single architecture.

Keywords: Medical Imaging; Transformers; Self-Supervised Learning; Multi-task Learning; 2D/3D Imaging

1. Introduction

Medical imaging spans a broad spectrum of acquisition methods, resolutions, and anatomies. Different tasks—such as detecting pulmonary lesions in X-rays, segmenting organs in CT scans, and classifying histopathological slides—have historically necessitated dedicated architectures optimized for each domain. While these specialized solutions can achieve strong results in narrow tasks, they fail to exploit latent similarities in tissue structures, imaging physics, and disease manifestations across imaging protocols [1]. Further complicating matters, many medical datasets are inherently small or imbalanced, as patient recruitment and expert annotations demand significant time and resources.

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1877-0509
10.1016/j.procs.2025.09.163

In an effort to unify learning across diverse modalities, we present **Medformer**: a *foundational transformer-based model* that can seamlessly handle 2D or 3D inputs, multiple clinical tasks (binary, multi-class, or ordinal classification), and various anatomical regions. The approach draws upon the concept of large vision transformers [2, 3] but introduces a key adaptation mechanism named *Adaptformer*. Specifically, we propose:

- **Input Adaptformer:** Incorporates domain-specific latent embeddings (e.g., 2D vs. 3D, modality type, anatomical region) to transform raw images or volumes into a standardized tokenized form that a single transformer backbone can process.
- **Output Adaptformer:** Learns task-specific embeddings, guiding how the backbone's universal representations become classification logits or other outputs, accommodating different label structures.

Our experiments on *MedMNIST* [4]—a curated set of 18 datasets ranging from chest X-ray classification to 3D abdominal CT labeling—indicate that Medformer performs on par with specialized models in both *single-task* and *multi-task* supervised modes. Additionally, a self-supervised (SSL) protocol (*VICReg* style [5]) substantially improves results, especially for tasks with minimal labeled data or subtle morphological details. By focusing on a single, adaptable backbone, we address challenges in parameter efficiency, data scarcity, and representational fragmentation across multiple medical imaging contexts.

2. Related Work

2.1. Self-Supervised Learning in Medical Imaging

Self-Supervised Learning (SSL) techniques reduce reliance on large annotated datasets by leveraging unlabeled data to learn robust features. Methods range from *context restoration* tasks (e.g., inpainting, jigsaw puzzles) to *contrastive* embeddings (SimCLR [6], MoCo [7]) and more sophisticated invariance-based frameworks (VICReg [5], Barlow Twins [8]). While SSL has been widely explored in natural images, its application in medical imaging remains largely domain-specific. For example, [9] uses 3D jigsaw tasks for volumetric scans, and [10] applies contrastive losses to chest X-ray images. However, combining *multiple* modalities (like 2D X-rays and 3D CT volumes) into a single SSL scheme is less common. We conducted a general overview of these research directions in previous work [11].

2.2. Transformers in Medical Imaging

The success of Vision Transformers (ViTs) [2] in computer vision has led to adaptations in medical imaging, particularly for segmentation and classification [3]. Transformers offer advantages such as global receptive fields and better modeling of long-range dependencies within a scan. Yet, existing transformer-based medical models typically handle *either* 2D images *or* 3D volumes, and often focus on a single clinical task. In contrast, Medformer is explicitly designed for *multi-task* and *multi-dimensional* input handling via its *Adaptformers*.

2.3. Unified Frameworks for Medical Imaging

Attempts at unified or multi-task models in medical imaging often rely on a shared Convolutional Neural Network (CNN) backbone plus multiple heads (e.g., for segmentation and classification). However, these methods usually restrict themselves to a single modality. Some works propose multi-dataset training [1], but do not systematically address input dimensionality mismatches or incorporate SSL. Recent transformer-based attempts such as **UniViTAR** [12] and **MedSAM++** [13] also pursue cross-modality unification, but differ from *Medformer* in two key ways. *(i)* None of them inject an explicit latent that distinguishes 2-D from 3-D inputs; instead, they treat volumetric data either as a stack of 2-D slices or omit it altogether. *(ii)* Their experiments remain focused on a single task family— *UniViTAR* benchmarks only 2-D image/video classification and semantic segmentation, whereas *MedSAM++* is evaluated exclusively on segmentation. Medformer stands out by *(a)* factoring domain knowledge (2D vs. 3D, modality, body-part) into small learned embeddings, and *(b)* applying the same main transformer layers for all datasets, thus forming a single, unified representation engine.

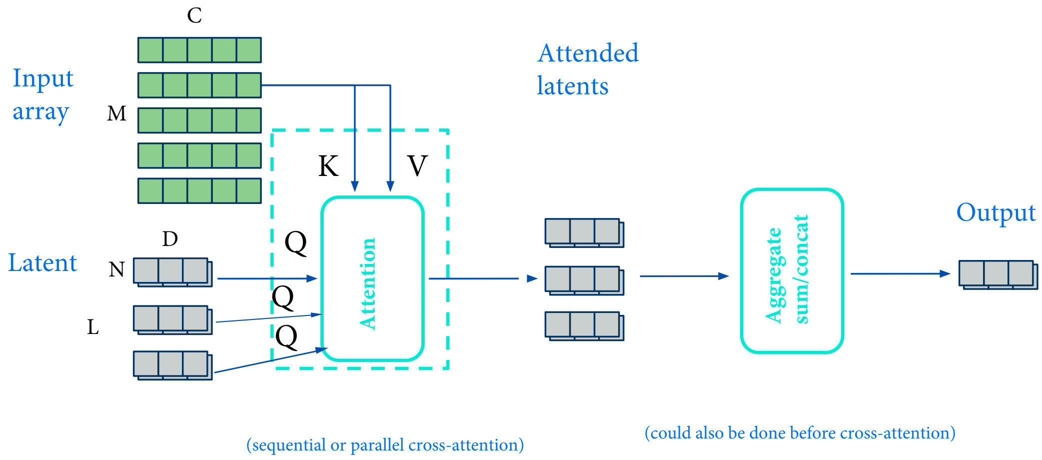

Fig. 1: Adaptformer Architecture

3. Methodology

3.1. Overall Architecture

Medformer consists of three principal parts:

1. **Input Adaptformer:** Takes a raw 2D image or 3D volume, creates patch tokens, and fuses them with learnable latent embeddings that represent the dimensionality, modality, and anatomical region.
2. **Main Body (Universal Transformer):** Processes these standardized tokens with a global self-attention mechanism. This backbone is *shared* across all tasks, promoting parameter reuse.
3. **Output Adaptformer:** Receives the high-level hidden representations, then cross-attends with *task-specific* latents to produce classification or regression outputs.

Rationale.. By separating domain-specific adaptation (Input Adaptformer) from task-specific adaptation (Output Adaptformer), Medformer ensures that the *same* large transformer backbone can handle different imaging shapes, modalities, and classification tasks. This design greatly reduces the duplication of parameters across specialized networks.

3.2. Input Adaptformer and Domain Latents

To handle the variability in medical imaging (2D vs. 3D, different modalities, and diverse anatomical regions), Medformer employs an **Input Adaptformer** that combines raw patches with a set of *domain-specific latent embeddings* through a cross-attention mechanism. Figure 1 provides a schematic view of this module.

Patchification and Position Embeddings.. Each image or volume x is first *patchified*. For a 2D input of size $H \times W$, we extract non-overlapping patches of size $p \times p$, producing a set of $N_{2D} = \frac{H \cdot W}{p^2}$ patches; for a 3D volume $D \times H \times W$, we similarly create $p \times p \times p$ subvolumes yielding $N_{3D} = \frac{D \cdot H \cdot W}{p^3}$ tokens. Each patch token is then augmented with the appropriate 2D or 3D positional embedding, ensuring that spatial or volumetric order is preserved.

Domain-Specific Latents.. We maintain three main categories of trainable latent embeddings:

- `dim_latents` (`2D_latent` or `3D_latent`) to signal whether the input is planar or volumetric.

- `modality_latents` (e.g., `xray`, `ct`, `microscopy`) to encode imaging characteristics (intensity ranges, contrast, etc.).
- `body_part_latents` (e.g., `chest`, `abdominal`, `brain`) to highlight the anatomical site and potential geometry or texture patterns typical of that region.

A given sample (e.g., a 2D chest X-ray) selects the correct embeddings (`2D_latent + xray + chest`) and concatenates them as a small set of latent tokens, L, which will be used in a cross-attention block.

Cross-Attention for Domain Conditioning.. Algorithm 1 summarizes the procedure for embedding an input image or volume. Essentially, the Input Adaptformer transforms the raw patch embeddings into domain-informed token representations by cross-attending with L:

Algorithm 1 Input Adaptformer (Simplified Pseudocode)

Require: x ▷ Raw image or volume (2D or 3D)
Require: $\ell_{\text{dim}}, \ell_{\text{mod}}, \ell_{\text{body}}$ ▷ Domain latents
$X_{\text{patch}} \leftarrow \text{Patchify}(x)$ ▷ Split into $p \times p$ (2D) or $p \times p \times p$ (3D) patches
$X_{\text{pos}} \leftarrow X_{\text{patch}} + \text{PosEmbed}(X_{\text{patch}})$ ▷ Add 2D or 3D position embeddings
$L \leftarrow \text{Concat}(\ell_{\text{dim}},\ \ell_{\text{mod}},\ \ell_{\text{body}})$ ▷ Shape: [#latent tokens, embedding dim]
for transformer block b in InputAdaptformerBlocks **do**
    $X_{\text{pos}} \leftarrow b(X_{\text{pos}},\ L)$ ▷ Cross-attention between patches and domain latents
end for
return X_{pos} ▷ Domain-adapted tokens

By using cross-attention, the patch tokens incorporate domain knowledge about dimensionality, modality, and anatomical region. This yields a more specialized token representation that is then fed to the main transformer backbone.

3.3. Main Body (Universal Transformer)

The central **Main Body** is a standard transformer encoder with L layers of multi-head self-attention and feed-forward blocks. It is deliberately *universal* and *task-agnostic*, receiving the domain-adapted tokens from the Input Adaptformer. Through global self-attention, this backbone captures common morphological or textural features across various domains and modalities. By decoupling domain adaptation from universal feature extraction, we avoid duplicating large numbers of parameters whenever a new modality or task is added.

3.4. Output Adaptformer and Task Latents

Once the universal transformer produces a set of hidden representations, Medformer uses an **Output Adaptformer** to generate task-specific predictions. Here, a different set of learnable *task latents* (e.g., classification in `chestmnist_binary`, ordinal regression in `retinamnist_ordinal`) tells the model how to interpret these hidden representations for the final output.

Task-Specific Cross-Attention.. The Output Adaptformer takes the hidden tokens H from the main transformer and cross-attends them with one or more task-latent tokens. If the task is, say, multi-class classification, there might be a single `task_latent` associated with that dataset's label structure. For a different dataset or label type (binary, ordinal), the relevant task embedding is activated.

Final Prediction Head.. After a few layers of cross-attention, a lightweight classifier head (often a linear or small MLP) translates the final pooled token embedding into logits or probability scores. Algorithm 2 gives a simplified outline:

This design allows a single universal representation to branch out for multiple label structures. If a new task arises (e.g., multi-label classification of pathologies in X-ray), we simply define a new task latent and a small output head, leaving the backbone and Input Adaptformer intact.

Algorithm 2 Output Adaptformer (Simplified Pseudocode)

Require: H ▷ Hidden tokens from universal transformer
Require: ℓ_{task} ▷ Task latent(s)
$H_{\text{cur}} \leftarrow H$ ▷ Initialize from backbone output
for transformer block b in OutputAdaptformerBlocks **do**
  $H_{\text{cur}} \leftarrow b(H_{\text{cur}}, \ell_{\text{task}})$ ▷ Cross-attention with task latent(s)
end for
$H_{\text{pooled}} \leftarrow \text{MeanPooling}(H_{\text{cur}})$
return ClassifierHead(H_{pooled})

Advantages..

- *Flexibility*: Adding tasks requires minimal overhead; we only introduce task latents and a final classification (or regression) head.
- *Param Efficiency*: The large universal transformer is reused across 2D/3D modalities, and the Input/Output Adaptformers remain relatively small modules guiding domain/task specialization.
- *Reduced Interference*: By "injecting" domain signals in the Input Adaptformer and focusing final interpretive steps in the Output Adaptformer, we mitigate catastrophic interference among tasks of different dimensionalities or label types.

These input-output adaptation modules thus form the key pillars of Medformer's capacity to unify disparate medical imaging tasks (X-ray vs. CT, binary vs. multi-class) under one high-capacity transformer model.

3.5. Self-Supervised Learning (SSL)

Medformer integrates SSL by adding a *self-supervised* head in the Output Adaptformer. We employ an invariance-based approach (e.g., VICReg) where each unlabeled image is augmented (random flips, rotations, slight intensity shifts), and the model learns to map these augmentations to similar embeddings. Figure 2 shows a typical SSL loss convergence plot. After SSL pretraining, we either:

- Fine-tune Medformer on a single labeled dataset for classification,
- Or continue multi-task supervised training, using the SSL-initialized weights for faster convergence.

Fig. 2: SSL Training Loss Curve. Reductions in invariance and variance terms are seen over epochs, improving the learned representation.

This workflow allows the model to benefit from large unlabeled cohorts, which is especially relevant given privacy constraints that limit label sharing but not necessarily the sharing of de-identified images.

3.6. Experiment Setup and Hyperparameters

Data:. We test on *MedMNIST* [4], a benchmark of 18 2D/3D datasets (Table 1). Each dataset covers different organs or tissue types, with tasks ranging from binary classification (e.g., pneumonia detection) to multi-class classification (e.g., organ labeling). The images are resampled to smaller resolutions (28×28 or 28^3) or higher resolutions (224×224 or 64^3). Figure 3 illustrates representative samples.

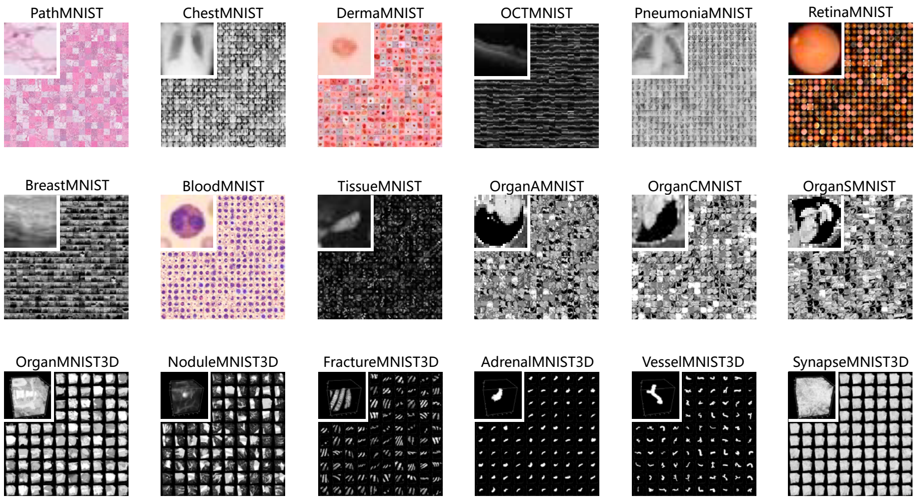

Fig. 3: MedMNIST: wide diversity in imaging modalities (X-ray, CT, microscopy, etc.) and tasks.

Table 1: MedMNIST dataset distribution. Each dataset contains 2D or 3D samples covering various body parts and classification tasks.

Dataset	**Modality**	**Dim**	**Task Type**	**#Samples**
PathMNIST	Histopathology	2D	Single-label	107,180
ChestMNIST	X-Ray	2D	Binary	112,120
DermaMNIST	Microscopy	2D	Single-label	10,015
RetinaMNIST	Fundus	2D	Ordinal	1,600
OrganMNIST3D	Abdominal CT	3D	Multi-class	1,742
NoduleMNIST3D	Chest CT	3D	Binary	1,633
FractureMNIST3D	Bone CT	3D	Single-label	1,370

Evaluation Metrics.. Accuracy (Acc) and area-under-the-ROC curve (AUC) are computed as

$$\text{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i], \quad \text{AUC} = \int_0^1 \text{TPR}(f)\, d(\text{FPR}),$$

where TPR and FPR are true and false-positive rates.

Training Regimes:. We compare:

- **Single-task Supervised**: Train Medformer on *one* dataset, selecting the appropriate domain latents and output head.
- **Multi-task Supervised**: Jointly train on multiple (or all) MedMNIST datasets, interleaving them in mini-batches. Each sample uses the correct dimensional, modality, body-part, and task latents.
- **Self-Supervised (SSL) Pretraining + Fine-tuning**: Pretrain with unlabeled images (or ignoring labels) using an invariance-based objective, then fine-tune on each dataset.

Hyperparameters:.

- Optimizer: `AdamW` [14], learning rate 10^{-3} or 10^{-4}, weight decay 10^{-4}.
- Batch size: 64 for 2D images; 16–32 for 3D volumes (memory constraints).
- Patch size: Typically 4 or 8 for 2D (depending on resolution), 2 or 4 for 3D volumes.
- Number of Transformer Layers: 4–6.
- Data Augmentation: RandAugment [15] (random flips, rotations, color jitter).

4. Results and Analysis

4.1. Single-Task Performance

We first assess Medformer on each dataset independently. As illustrated in Table 2 (2D tasks) and Table 3 (3D tasks), single-task supervised training with domain-appropriate latents yields strong results, often on par with specialized CNNs or Vision Transformers reported in the literature. Notably:

- **PathMNIST**: Achieves near 97% accuracy and nearly perfect AUC, matching prior state-of-the-art on this histopathology classification.
- **ChestMNIST, PneumoniaMNIST**: Both based on X-ray images, see classification metrics consistent with the official MedMNIST baselines.
- **NoduleMNIST3D, OrganMNIST3D**: Medformer's 3D patchification effectively detects nodules or classifies organs, aligning with specialized 3D CNN performance.

Table 2: Performance of Medformer on select 2D classification tasks (small resolution).

Dataset	Task Type	AUC	Accuracy (%)
PathMNIST	Multi-class	0.999	97.2
ChestMNIST	Binary	0.525	90.0
DermaMNIST	Multi-class	0.772	66.8
PneumoniaMNIST	Binary	0.986	94.2

Table 3: Performance of Medformer on select 3D classification tasks (small resolution).

Dataset	Task Type	AUC	Accuracy (%)
OrganMNIST3D	Multi-class	0.991	93.7
NoduleMNIST3D	Binary	0.877	84.2
FractureMNIST3D	Single-label	0.536	42.7

Some tasks are inherently more challenging (e.g., *FractureMNIST3D* has subtle bone features), which results in lower absolute accuracy. However, these single-task experiments confirm that Medformer, despite its general-purpose design, can function competitively *within* each domain.

4.2. Multi-Task Learning

We next jointly train Medformer on multiple datasets, mixing 2D and 3D data in the same mini-batches. Table 4 shows that *most* tasks maintain or exceed single-task performance, suggesting that the shared backbone can learn generic morphological cues without suffering catastrophic interference. In particular:

- **DermaMNIST** (microscopy) benefits from multi-task synergy, indicating that morphological features from other microscopic tasks (e.g., *PathMNIST*) transfer effectively.
- **ChestMNIST** and **PneumoniaMNIST** cross-benefit from both being X-ray datasets of the chest region.
- **3D tasks** like *OrganMNIST3D* often improve if the other 3D scans share a similar body region (e.g., abdominal CT).

Table 4: Multi-task learning performance by modality and dimension. Each dataset uses the relevant domain latents for the Input Adaptformer.

Dataset	Modality	Dim	AUC	Acc (%)
PathMNIST	Microscopy	2D	0.995	96.2
DermaMNIST	Microscopy	2D	0.203	89.8
ChestMNIST	X-ray	2D	0.717	89.9
OrganMNIST3D	CT	3D	0.978	73.2
NoduleMNIST3D	CT	3D	0.862	86.0

Although multi-task training occasionally yields minor trade-offs (e.g., some tasks see small dips in AUC but higher accuracy), the net effect is generally neutral or beneficial. This result implies that domain embeddings effectively isolate unique modalities/dimensions while still leveraging broad morphological patterns in the shared transformer body.

4.3. Self-Supervised Pretraining and Fine-Tuning

Table 5 presents fine-tuning results after SSL pretraining on unlabeled images. We observe:

- **Faster Convergence:** Fine-tuning consistently requires fewer epochs to reach comparable accuracy vs. from-scratch supervised runs.
- **Performance Gains:** Data-scarce sets (e.g., *DermaMNIST*, *BreastMNIST*) benefit the most, often showing higher AUC or improved sensitivity to minority classes.
- **Limited Gains on Large Sets:** Datasets like *PathMNIST*, which already have over 100k samples, see minimal improvement from SSL, consistent with prior findings that SSL is most impactful when labeled data are limited.

Table 5: Example single-task fine-tuning performance from SSL pretraining.

Dataset	Modality	Dim	AUC	Accuracy (%)
PneumoniaMNIST	X-ray	2D	0.986	94.2
BloodMNIST	Microscopy	2D	0.994	93.2
OrganMNIST3D	CT	3D	0.986	87.5
BreastMNIST	Ultrasound	2D	0.850	75.6

Finally, Table 6 compares single-task (ST), multi-task (MT), and SSL fine-tuning for select datasets. Small or imbalanced datasets often see double-digit accuracy gains when moving from single-task to SSL or multi-task. Conversely, large tasks with ample labels only show marginal gains.

Table 6: Comparison of ST, MT, and SSL across MedMNIST (select examples).

Dataset	ST AUC	MT AUC	SSL AUC	Observation
PathMNIST	0.999	0.995	0.998	Large dataset, minor SSL gain
DermaMNIST	0.772	0.203	0.866	SSL helps data-scarce domain
ChestMNIST	0.525	0.717	0.703	Notable multi-task synergy

4.4. Scaling to Larger Resolutions

While all experiments above used 28×28 or $28 \times 28 \times 28$ inputs, we tested a *large* variant on 224×224 (2D) and $64 \times 64 \times 64$ (3D). On tasks requiring fine detail (e.g., *DermaMNIST*, *BloodMNIST*), we observed moderate performance improvements. However, tasks with mostly coarse structures or small data did not always benefit, possibly due to overfitting on more extensive patch embeddings. Moreover, memory usage increases substantially (from 10GB of VRAM usage to 32GB for the multi-task training regime) suggesting that domain-specific preprocessing (e.g., denoising, partial volume cropping) could be crucial for stable training. Hence, while the architecture scales conceptually, practical adoption may require carefully tuned hyperparameters for high-resolution volumetric data.

5. Discussion

Our extended results confirm the viability of a **universal transformer** that addresses 2D or 3D medical imaging tasks by injecting domain and task information at the periphery. Several themes emerged:

Domain Embeddings Facilitate Multi-Modal Coverage.. By encoding dimensionality (2D vs. 3D), modality (CT, X-ray, etc.), and anatomical location, the Input Adaptformer successfully shapes raw patches into feature tokens that a single backbone can handle. This avoids catastrophic interference, as indicated by stable multi-task performance.

Self-Supervised Benefits in Low-Data Regimes.. Consistent with earlier SSL insights [16, 9], unlabeled pretraining yields significant gains on smaller datasets like *BreastMNIST* or *FractureMNIST3D*. The model effectively learns to represent morphological attributes in unlabeled scans, reducing the labeled sample burden.

Trade-Offs at High Resolution.. While scaling to larger input sizes can capture finer details, it introduces memory and overfitting challenges. Future directions might incorporate domain-tailored preprocessing or advanced augmentations for 3D volumes to curb overfitting and exploit subtle structural cues more effectively.

Toward Realistic Clinical Integration.. A single model that can flexibly switch among tasks and modalities is appealing in hospital environments, where multiple classification (or segmentation) routines might be needed in parallel. In principle, institutions with large unlabeled archives could exploit SSL to refine a universal backbone, then fine-tune specific heads for local tasks. Nonetheless, additional steps—like interpretability modules, domain adaptation for scanner-specific artifacts, and evaluation on prospective clinical data—remain crucial before widespread deployment.

6. Conclusion

In this paper, we presented **Medformer**, a transformer-based model that unifies multiple modalities, anatomies, and classification tasks under one adaptable framework. Our experimentation on the diverse MedMNIST suite underscores several strengths:

- **Universal Transformer Core:** By separating domain adaptation (Input Adaptformer) and task adaptation (Output Adaptformer) from a shared transformer, Medformer avoids duplicating large parameter sets for each dataset.
- **Domain-Specific Embeddings:** Small learnable latents (for dimensionality, modality, body-part) effectively guide patch tokens, preserving each dataset's uniqueness while supporting parameter sharing.
- **Improved Efficiency through SSL and Multi-tasking:** Tasks with limited labeled data show marked gains when leveraging unlabeled data or training alongside larger tasks.
- **No Catastrophic Interference:** Even when training 2D and 3D tasks concurrently, performance remains stable or improves, indicating that Medformer's domain latents help isolate distinct modalities.

Limitations.. While Medformer handles varied resolutions and imaging types, high-resolution 3D volumes can still pose computational challenges. Domain-specific preprocessing (e.g., intensity normalization, region-of-interest cropping) may be essential to fully exploit the model's capacity. Also, datasets with highly specialized domains (e.g., rare pathologies) might require additional domain-tailored embeddings.

Future Work.. We aim to extend Medformer with more specialized augmentations and advanced attention mechanisms for 3D. The approach can also scale to larger repositories of unlabeled medical scans, further enhancing representation learning through SSL. Applying Medformer to segmentation or detection tasks, potentially via specialized Output Adaptformers, is another promising direction.

Overall, this study demonstrates that a single, transformer-based model can effectively unify and handle the diversity in medical imaging across modalities, dimensionalities, and label structures. Medformer provides a practical stepping stone toward more comprehensive foundation models in medical AI, reducing fragmentation and improving transferability across datasets.

References

[1] V. Cheplygina, M. De Bruijne, J. P. Pluim, Not-so-supervised: a survey of semi-supervised, multi-instance, and transfer learning in medical image analysis, Medical image analysis 54 (2019) 280–296.

[2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., An image is worth 16x16 words: Transformers for image recognition at scale, arXiv preprint arXiv:2010.11929 (2020).

[3] J. Chen, Y. Lu, Q. Yu, X. Luo, E. Adeli, Y. Wang, L. Lu, A. L. Yuille, Y. Zhou, Transunet: Transformers make strong encoders for medical image segmentation, arXiv preprint arXiv:2102.04306 (2021).

[4] J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, B. Ni, Medmnist v2-a large-scale lightweight benchmark for 2d and 3d biomedical image classification, Scientific Data 10 (1) (2023) 41.

[5] A. Bardes, J. Ponce, Y. LeCun, Vicreg: Variance-invariance-covariance regularization for self-supervised learning, arXiv preprint arXiv:2105.04906 (2021).

[6] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, A simple framework for contrastive learning of visual representations, in: International conference on machine learning, PMLR, 2020, pp. 1597–1607.

[7] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, Momentum contrast for unsupervised visual representation learning, in: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2020, pp. 9729–9738.

[8] J. Zbontar, L. Jing, I. Misra, Y. LeCun, S. Deny, Barlow twins: Self-supervised learning via redundancy reduction, in: International Conference on Machine Learning, PMLR, 2021, pp. 12310–12320.

[9] A. Taleb, W. Loetzsch, N. Danz, J. Severin, T. Gaertner, B. Bergner, C. Lippert, 3d self-supervised methods for medical imaging, Advances in neural information processing systems 33 (2020) 18158–18172.

[10] R. Goel, U. Nath, Y. Wang, A. C. Silva, T. Wu, Y. Yang, Learning low-rank feature for thorax disease classification, arXiv preprint arXiv:2404.18933 (2024).

[11] C. Simionescu, A. Iftene, Deep learning research directions in medical imaging, Mathematics 10 (23) (2022) 4472.

[12] L. Qiao, Y. Gan, B. Wang, J. Qin, S. Xu, S. Yang, L. Ma, Univitar: Unified vision transformer with native resolution, arXiv preprint arXiv:2504.01792 (2025).

[13] J. Ma, Y. He, F. Li, L. Han, C. You, B. Wang, Segment anything in medical images, Nature Communications 15 (1) (2024) 654.

[14] I. Loshchilov, F. Hutter, Decoupled weight decay regularization, arXiv preprint arXiv:1711.05101 (2017).

[15] E. D. Cubuk, B. Zoph, J. Shlens, Q. V. Le, Randaugment: Practical automated data augmentation with a reduced search space, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2020, pp. 702–703.

[16] X. Zhuang, Y. Li, Y. Hu, K. Ma, Y. Yang, Y. Zheng, Self-supervised feature learning for 3d medical images by playing a rubik's cube, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, Springer, 2019, pp. 420–428.